

Task Type	Guided Hands-on Coding
Description of the Task	Image files and prepared Jupyter Notebook (.ipynb) resources are shared at the beginning of the session. You follow the notebook and implement each transformation while observing the visual effects. Task 1.1: Point Operations Playground You apply basic point transformations to a grayscale image. Implement image negative ($s = 255 - r$) using NumPy and verify with OpenCV. Apply log transformation to enhance dark regions. Perform gamma correction using $\gamma = 0.4$, $\gamma = 1.0$, and $\gamma = 2.5$ and display the outputs next to the original image. Outcome: Observe how intensity transformations affect brightness and contrast. Task 1.2: Histogram Detective You load different types of images (overexposed, underexposed, high contrast, normal). Compute and visualize histograms for each image and analyze how histogram shape reflects brightness distribution and contrast. Outcome: Interpret histogram patterns and relate them to image quality. Task 1.3: Histogram Equalization You first perform histogram equalization manually using a small matrix example (count frequencies, compute cumulative distribution, scale values, remap pixels). Then apply <code>cv2.equalizeHist()</code> on a real image and compare before/after results and histograms. Outcome: Understand how histogram equalization improves global contrast. Task 1.4: Thresholding & Intensity Slicing You apply binary thresholding using <code>cv2.threshold()</code> and test multiple threshold values to observe changes in segmentation. Perform intensity slicing by isolating a specific pixel range and suppressing all other values. Outcome: Understand threshold sensitivity and basic segmentation techniques. Task 1.5: Histogram Equalization vs CLAHE You apply global histogram equalization and CLAHE (<code>cv2.createCLAHE()</code>) on the same image. Compare the original image, equalized image, and CLAHE result along with their histograms. Outcome: Observe the limitations of global equalization and how CLAHE preserves local contrast. Task 1.6: CNN Sensitivity to Intensity Transformations You run the same image through a pretrained CNN (e.g., ResNet) multiple times using

	different intensity transformations: original image, gamma-darkened, gamma-brightened, histogram-equalized, and CLAHE. Record the predicted class and confidence score for each version. Outcome: Understand how image intensity and preprocessing influence CNN predictions and confidence.
Time Allocation	Class Timing Summary (2 Hours): - First 10 min (Concept recap and workshop overview) - Next 30 min (Task 1.1 – Point operations) - Next 25 min (Task 1.2 – Histogram analysis) - Next 25 min (Task 1.3 – Histogram equalization) - Next 20 min (Task 1.4 – Thresholding & intensity slicing) - Next 20 min (Task 1.5 – CLAHE comparison) - Final 20 min (Task 1.6 – CNN experiment and discussion)
Submission of Work	Not Applicable. You demonstrate results during the session by running notebook cells and showing visual outputs. The focus is on experimentation and conceptual understanding of transformations.